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Russian Geology and Geophysics 59 (2018) 1204–1216

RUSSIAN GEOLOGY AND GEOPHYSICS

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Basites of the Vilyui paleorift: geochemistry and sequence of intrusive events

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Received 15 January 2018; received in revised form 24 April 2018; accepted 15 June 2018

Abstract

We present results of study of the basites of the Vilyui paleorift. Their $^{40}Ar/^{39}Ar$ isotopic ages were used to establish the sequence of their intrusion, which lasted ~18 Myr. Dolerite sills of the Kontai–Dzherba zone were the first crystallizing phase (378.0–380.7 Ma), which was followed by the intrusion of dikes of the Vilyui–Markha zone (372–373 Ma). Dikes of the Chara–Sinsk zone were the last crystalizing phase (362–364 Ma). The prolonged plume–lithosphere interaction resulted in the successive enrichment of basaltic melt with titanium, phosphorus, REE, HSFE, and LILE. Accumulation of the melt and formation of a magmatic lens at the crust–mantle boundary led to early crystallization of the melt at a depth of 35–40 km from the paleosurface under reducing conditions at a temperature of 1450–1470 ºC. We assume that the inception and sequent opening of the Vilyui paleorift were associated with the relative rotation of 22° of the Angara–Anabar block with respect to the Aldan block in Early Famennian time. The Euler pole of rotation was located in the southern part of the paleorift, and a melting zone was directly beneath it.
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Keywords: Middle Paleozoic; $^{40}Ar/^{39}Ar$ geochronology; paleomagnetic studies; dike swarms; dolerites

Introduction

The Middle Paleozoic Vilyui paleorift is a major structural element of the eastern Siberian platform. Its rift origin was first substantiated by Levashov (1975). The rift structure was discussed in detail by Gaiduk (1988), and the history of rift formation was reconstructed using the backstripping method by Polyansky et al. (2013). Data on the tectonic structure, crustal structure revealed by deep seismic sounding, the wide occurrence of alkaline mafic magmatism and the presence of multiple dikes (Grachev, 1977) unequivocally point to the rifting origin of this structure. Structurally, the depressions in the rift basin are NE-oriented grabens and half-grabens separated by uplifts (horsts) and bounded by faults with amplitudes reaching 1.5 km. Paleomagnetic data (Pavlov et al., 2008; Tomshin and Konstantinov, 2005) indicate a relative rotation of 20°–25° of the Anabar–Angara block with respect to the Aldan block. This rotation resulted in heterogeneous crustal extension with the formation of a depression, when crust in the western and eastern parts of the rift was reduced in thickness by a factor of 1.55 and 2.15, respectively. Gaiduk (1988) established several stages in the evolution of the Vilyui rift system: the Middle Devonian stage was marked by the formation of a gentle depression and carbonate sedimentation in Givetian time; Early Famennian stage was marked by uplift and massive intrusion of mafic magma; Famennian–Early Carboniferous stage was marked by the inception of rift structures, grabens, horsts accompanied by predominantly felsic volcanism and evaporite deposition. New rift-bounding faults were generated and acted as the main magma conduits, including the Vilyui–Markha fault in the northwest, the Chara–Sinsk fault in the southeast, and Kontai–Dzherba fault in the inward-dipping termination of the rift structure (Fig. 1). Mafic magmas were intruded along these rift-bounding faults forming extensive dike swarms of the same name, which point to the rift origin of the Vilyui structure. Dikes are far more prevalent across the Vilyui–Markha and Chara–Sinsk zones, while sills predominate in the Kontai–Dzherba zone, and most of them were injected into Lower Paleozoic carbonate strata. The identified fault zones have a northeast strike.

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http://dx.doi.org/10.1016/j.rgg.2018.09.002

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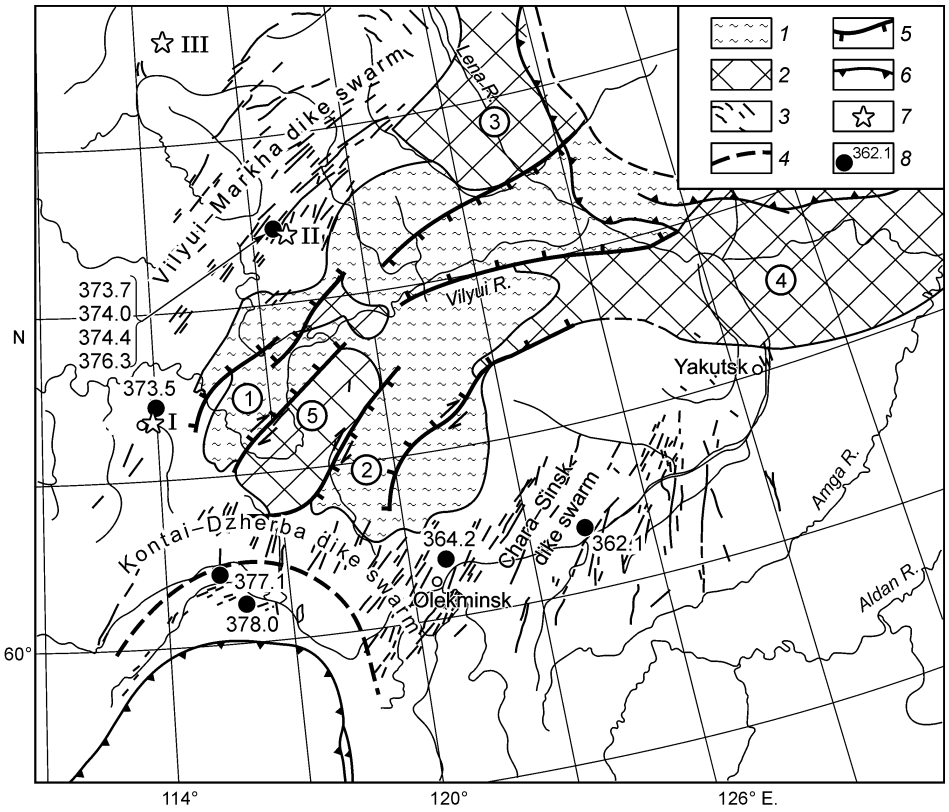


Fig. 1. Schematic structure of the Vilyui paleorift (Tomshin, 2000). *1*, depressions willed with terrigenous sediments, up to 6 km thick (1, Ygyatta; 2, Kempendyai); *2*, major uplifts (3, Dzhardzhan; 4, Yakutsk; 5, Suntar); *3*, dike swarms; *4*, sills of the Kontai–Dzherba swarm; *5*, faults; 6, fronts; *7*, kimberlite fields (I, Mirnyi; II, Nakyn; III, Daldyn–Alakit); *8*, age, Ma.

The Kontai–Dzherba field of dikes and sills, 250 km long and about 85 km wide, is located at the southwestern margin of the rift. The Vilyui–Markha dike swarm extends for almost 700 km ranging in width from 30 km in the south to 150 km in the north. In the northern part, it changes its strike from mostly NE to E direction with the formation of the Chimidikyan swarm of en-echelon dikes. The Chara–Sinsk dike swarm extends for 450–500 and varies in width from 170 km in the north to 290 km in the south.

Compositional features of mafic intrusions in the Vilyui paleorift

The Kontai–Dzherba swarm of dikes and sills. The dolerite sills are mainly found in the Kontai–Dzherba zone, while dike-like bodies are less common. The sills appear to have been intruded into horizontal or shallowly dipping Lower Paleozoic strata, the vertical section of which contains up to 10 levels of sill intrusion. The thickness of individual sills varies from 15 to 100–110 m. The greatest thicknesses of intrusions are found in the lower reaches of the Bol. Patom River (Talaktakh sill, up to 100 m thick) and along the Lena River (from the Kontai–Dzherba River to the Tochilnaya River). In the northern direction, the thickness of sills decreases to <40 m as the degree of deformation increases. Intrachamber magma fractionation processes gave rise to a variety of rock types, ranging in composition from ophitic dolerites and prismatic-ophitic gabbro-dolerites to troctolitic gabbro-dolerites containing up to 25% olivine. Some thick sills may contain schlieren of ferrogabbro and syenogabbro. All rock varieties contain from 3 to 10% porphyritic plagioclase, indicative of crystallization of basaltic magma. The contact facies of the sills are composed of porphyritic microdolerite with plagioclase and olivine phenocrysts. The rocks of the intrusions are prismatic-ophitic gabbro-dolerite consisting essentially of plagioclase ($An_{72–38}$), augite ($Wo_{40–43}En_{46–40}Fs_{18–15}$), olivine ($Fa_{35–23}$) and oxide ore minerals. The mesostasis can include quartz, K-feldspar (sometimes as a micropegmatitic intergrowth), hornblende, biotite, and apatite.

The Vilyui–Markha dike swarm. Magmatic activity in this dike swarm was associated with the intrusion of sills, chonoliths, dikes and eruptive vents of basic composition. All of them are intruded into sedimentary carbonate rocks of Cambrian and Ordovician age. The dikes are a few tens of kilometers long and range in thickness varies from 6–8 to 80 m. The thickness of the sills may reach only a few tens of meters, as indicated by drilling data. Late-stage basaltic lavas of the Appainskaya and Emyaksinskaya Formations at the southwestern edge of the dike swarm are associated with the formation of the Vilyui paleorift valley and may be only spatially related to the swarm of dolerite dikes.

The following types of rocks are recognized based on their compositional character, which recorded the main evolutionary

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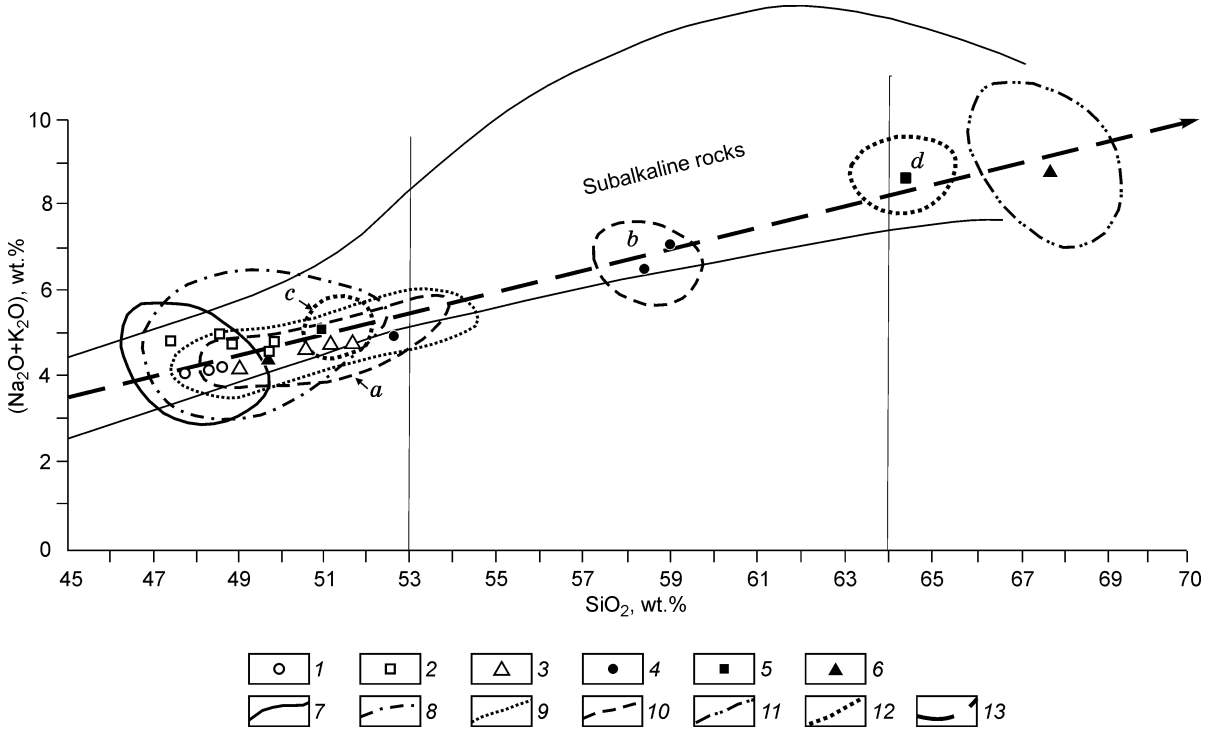


Fig. 2. Average compositions and fields for different of intrusive bodies of the Chara–Sinsk fault zone (Tomshin et al., 1985). *1–6*, average compositions (*1*, dolerite and gabbro-dolerite; *2*, quartz gabbro-dolerite; *3*, subalkaline gabbro-dolerite; *4*, quartz monzonite porphyry of two-phase dikes; *5*, quartz syenite porphyry of two-phase dikes; *6*, quartz syenite porphyry of the Olekma intrusion); *7–12*, fields for the individual analyses of various intrusion groups: *7*, dolerite and gabbro-dolerite; *8*, quartz gabbro-dolerite; *9*, subalkaline quartz gabbro-dolerite in single-phase dikes; *10*, two-phase dikes (*a*, subalkaline quartz gabbro-dolerite; *b*, quartz monzonite porphyry); *11*, quartz syenite porphyry of the Olekma intrusion; *12*, two-phase dikes (*c*, subalkaline quartz gabbro-dolerite, *d*, quartz syenite porphyry); *13,* differentiation trend for the subalkaline tholeiitic-basaltic magma of the Chara–Sinsk fault zone.

trends of the basaltic magma produced by prechamber and intrachamber processes: ophitic and prismatic-ophitic gabbro-dolerite, dikes of spotted and poikiloophitic gabbro-dolerites, intrusions of taxitoophitic gabbro-dolerite, fissure intrusions of prismatic-ophitic gabbro-dolerite containing local segregations of anorthositic gabbro-dolerite or monzonite porphyry, dikes of subalkaline gabbro-dolerite, tuff-breccia pipes. All gabbro-dolerites contain quartz.

Intrusions are characterized by poorly developed processes of intrachamber magma differentiation, as indicated by the presence of a single rock variety in each particular magmatic body. The rare exception is some magmatic bodies that were formed by melts experienced a protracted prechamber evolution at great depths along an anorthositic (Oleinikov and Tomshin, 1976) or monzonitic (Korolev, 1991; Tomshin et al., 1985) differentiation trend. At the present time, segregations of anorthositic gabbro-dolerites and monzonite porphyries have been reported only in dikes of the Nakyn kimberlite field (Zemnukhov et al., 2005). The plagiodolerite dikes of the Chimidikyan swarm are found only locally in the ramp zone of the northwestern margin of the Vilyui–Markha swarm. A peculiar compositional feature of the plagiodolerite is the presence of abundant (up to 70 vol.%) small porphyritic segregations of plagioclase of labradorite composition ($An_{71–63}$). Besides plagioclase, other phenocrysts in these rocks include Mg-rich olivine and octahedra of chromian (up to 1.2% Cr) titanomagnetite.

The dominant type of intrusive rocks of the Vilyui–Markha dike swarm belt is prismatic-ophitic gabbro-dolerite. A distinctive feature of these rocks is the presence of quartz, a ubiquitous late-magmatic phase, which may account for 5–6 vol.%. Most gabbro-dolerites contain 3–5% of early crystallizing phases such as bytownite ($An_{88–80}$), chrysolite–hyalosiderite ($Fa_{21–35}$) and less common magnesian clinopyroxene ($Wo_{35–37}En_{54–52}Fs_{11–12}$). The anorthositic gabbro-dolerite usually contain the largest volume proportion of early plagioclase (18–25 vol.%). The intrachamber mineral paragenesis of gabbro-dolerites consists of plagioclase of a more felsic composition ($An_{78–44}$), augite ($Wo_{43–37}En_{49–41}Fs_{15–21}$), hortonolite ($Fa_{39–56}$), oxide ore minerals, and apatite. Other minerals common to Middle Paleozoic basites are hornblende, biotite, and K-feldspar. The fraction of the last two minerals together with quartz may reach 15 vol.%.

The Chara–Sinsk dike swarm. The onset of magmatic activity in this swarm was associated with the intrusion of sills and dikes of gabbro-dolerite and their quartz analogues. This stage was followed by the emplacement of subalkaline quartz gabbro-dolerites, which were replaced by quartz monzonite porphyries. The late stage of magmatic activity was represented by quartz syenite porphyries occurring as individual intrusions (Olekma stock) or later multiple dikes (Tomshin et al., 1985). The southern swarm margin is dominated by the generation of composite multiphase bodies of "dike-in-dike" type, with the average thickness of individual dikes of >200 m.

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Table 1. Average whole-rock composition (wt.%) of Middle Paleozoic basites of the Vilyui paleorift

Component	Fault zone															
	Kontai–Dzherba					Vilyui–Markha										
	M-27*	M-28	M-44	M-32	M-41	MIR-12	OB-130	OB-233	OB-288	OB-251	38/43	OB-242	50/49-17	OB-501	OB-510	OB-528
SiO_2	49.20	49.21	49.28	49.35	48.82	46.86	48.80	48.26	48.71	48.60	47.13	47.49	49.03	48.31	48.76	48.40
TiO_2	2.01	2.09	2.30	2.40	2.04	3.12	3.20	2.32	2.81	2.33	3.55	3.78	2.93	2.62	2.94	3.06
Al_2O_3	15.20	13.85	13.84	14.32	14.15	12.64	13.46	13.96	13.85	14.50	13.77	13.03	14.33	13.13	12.32	13.89
Fe_2O_3	4.43	3.19	4.06	4.70	3.28	7.15	6.05	2.59	2.57	4.88	6.21	4.85	4.09	3.10	6.32	3.22
FeO	7.79	10.02	9.68	8.46	10.15	7.07	8.94	11.50	10.39	8.20	8.91	10.59	8.56	11.69	9.44	9.80
MnO	0.17	0.18	0.19	0.19	0.21	0.21	0.19	0.20	0.23	0.16	0.20	0.12	0.17	0.18	0.15	0.22
MgO	5.40	6.60	6.05	6.25	6.38	5.52	5.01	5.88	5.97	5.65	6.03	5.57	5.64	6.46	5.58	6.25
CaO	9.54	10.71	10.29	9.66	10.35	9.63	8.95	10.00	10.16	9.59	8.63	9.02	9.37	10.21	9.23	9.21
Na_2O	2.13	2.20	2.37	2.31	2.39	3.27	2.09	2.21	2.46	2.49	2.06	2.09	2.73	2.24	2.66	2.61
K_2O	1.10	0.82	0.86	0.94	0.76	1.58	1.27	1.04	0.98	1.32	1.67	1.49	1.62	0.97	1.15	1.33
P_2O_5	0.25	0.24	0.27	0.31	0.23	0.33	0.60	0.31	0.33	0.29	0.35	0.52	0.33	0.29	0.41	0.35
H_2O	1.64	1.33	1.22	1.33	1.38	1.76	1.32	1.21	1.07	0.73	1.23	1.73	1.35	1.59	1.34	2.10
Total	99.18	100.5	100.4	100.3	100.1	99.54	99.89	99.58	99.53	100.0	99.73	100.2	100.1	100.4	100.3	100.5
Mg#	45	44	45	47	46	42	38	43	46	44	43	40	45	44	40	47
n	4	7	5	8	13	3	8	5	9	22	9	10	4	23	6	27

Component	Fault zone															
	Vilyui–Markha			Chara–Sinsk												
	OB-566	OB-569	OB-571	M-6	M-11	M-14	Tb-14	M-19	M-24	M-26	M-56	M-57	M-69	M-70	M-71	M-115
SiO_2	47.77	48.37	47.42	49.62	47.78	49.27	49.74	48.72	48.94	51.62	47.82	48.23	51.21	47.32	50.51	47.65
TiO_2	3.62	3.16	3.37	3.52	3.71	2.38	2.99	3.80	3.37	2.84	3.31	3.56	3.13	3.81	3.16	3.47
Al_2O_3	13.7	14.62	14.15	12.9	13.60	15.82	14.67	13.54	12.73	12.96	14.35	13.94	12.92	13.80	12.70	13.18
Fe_2O_3	6.35	4.4	5.14	6.85	5.59	4.03	5.41	5.27	7.24	6.64	5.79	4.57	7.81	4.83	6.19	7.15
FeO	8.28	9.48	8.84	9.09	8.98	7.93	8.74	9.31	8.04	7.81	7.81	9.48	6.86	9.03	8.52	7.02
MnO	0.18	0.18	0.21	0.2	0.17	0.15	0.18	0.17	0.19	0.19	0.17	0.18	0.28	0.17	0.24	0.20
MgO	4.36	4.61	5.45	4.63	5.21	4.95	3.74	4.58	4.65	4.56	5.39	5.26	4.08	5.98	4.66	6.31
CaO	9.49	9.12	9.05	7.34	9.13	10.29	8.75	8.48	8.41	6.57	8.90	8.44	6.55	7.69	6.95	8.67
Na_2O	2.39	2.16	2.39	2.46	2.80	2.71	2.87	2.49	3.13	3.19	3.00	2.94	3.75	2.57	3.09	2.57
K_2O	1.22	1.15	1.26	1.91	2.12	0.70	1.92	2.07	1.02	1.44	1.00	1.16	0.97	2.12	1.52	1.55
P_2O_5	0.4	0.34	0.42	0.63	0.58	0.25	0.45	0.59	0.56	0.55	0.40	0.42	0.54	0.43	0.52	0.34
H_2O	1.36	1.56	1.44	0.84	0.47	1.18	0.59	0.65	1.73	1.77	2.04	1.84	1.73	2.14	2.18	2.13
Total	99.12	99.15	99.16	99.99	100.2	99.66	100.1	99.67	100.0	100.2	100.0	100.0	99.82	99.87	100.24	104.3
Mg#	36	38	42	35	40	43	33	37	36	37	42	41	34	44	37	46
n	5	5	12	7	8	10	17	9	11	14	13	8	12	8	11	5

Note. Major oxides were analyzed by wet chemical technique at the Laboratory of Physicochemical Analytical Methods (Geology of Diamond and Precious Metals Institute, Siberian Branch, Russian Academy of Sciences); $Mg\# = Mg^{2+} \times 100/(Mg^{2+} + 0.85 \times Fe^{2+})$; *n*, the number of analyses.

To the north, multiple dikes change into laterally extensive dikes (e.g., the Tuolba dike, 250 km long and up to 120 m thick). The occurrence of multiple dikes comprising two to three differentiates with different content of silica and alkalis (e.g., the early gabbroic differentiates and late monzonitic differentiates) suggests that they could have been produced during the evolution of a single parental magma of subalkaline tholeiitic basalt nature (Fig. 2). This differentiation was discussed in previous studies (Oleinikov et al., 1984) and described as a monzonitoid trend (Koroleva, 1991; Tomshin et al., 1985) representing differentiation of a tholeiitic basaltic melt. Quartz gabbro-dolerites of the Chara–Sinsk swarm are also found as separate bodies and represent the early-crystallizing phases in composite dikes. The gabbro-dolerites are characterized by the constant content of early plagioclase (from 3–5 to 15%). In addition to early plagioclase ($An_{75–68}$), other phases that crystallized at the prechamber stage include chrysolite ($Fa_{21–24}$), magnesian augite ($Wo_{37}En_{52}Fs_{11}$), almandine and spessartine–almandine garnet, rutile, and moissanite. The phases representing the intrachamber stage of crystallization are more felsic plagioclase ($An_{58–33}$), more Fe-rich augite ($Wo_{36–38}En_{43}Fs_{21–19}$) and olivine ($Fa_{35–54}$), as well as hornblende, biotite, and micropegmatitic intergrowth of quartz and K-feldspar.

Whole-rock major and trace element geochemistry of basites

Major oxide and trace element composition of the basites of the Vilyui paleorift is given in Tables 1 and 2. In this study,

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Table 2. The content of petrogenic (wt.%) and rare (ppm) elements in representative basite samples from the Vilyui rift

Component	Fault zone																
	Kontai–Dzherba			Vilyui–Markha								Chara–Sinsk					
	M-28-5	M-43-4	M-134-6	MIR12/24	130/D-1	T-68-60	OB-233-1	OB-802-7	38/43-91	OB-566-1	OB-534-4	M-14e	M-18-2	M-73-1	M-72-1	M-26-1	TB-14-9
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
SiO_2	49.17	48.76	49.48	47.56	48.59	49.2	48.13	47.39	48.23	47.96	48.84	50.57	49.47	49.52	49.56	52.95	49.44
TiO_2	2.40	1.73	2.06	3.26	3.57	2.48	2.53	4.01	2.6	3.51	3.11	2.18	3.55	3.17	3.38	2.64	3.53
Al_2O_3	13.7	13.62	15.55	12.87	13.08	15.38	14.24	12.88	14.46	14.08	13.05	17.33	12.4	12.9	12.36	13.26	13.15
Fe_2O_3	4.53	2.46	3.08	6.64	4.92	3.43	2.79	6.92	2.94	5.97	3.38	3.77	3.67	6.5	6.97	5.84	5.19
FeO	9.71	9.69	8.34	6.75	10.08	9.42	10.25	9.63	9.82	8.46	12.44	7.67	8.7	8.22	9.06	8.46	9.31
MnO	0.19	0.2	0.16	0.2	0.2	0.17	0.19	0.23	0.2	0.12	0.25	0.15	0.14	0.18	0.26	0.23	0.2
MgO	6.15	5.83	5.63	5.5	5.18	5.04	5.57	5.36	5.12	3.87	4.97	3.65	4.8	4.51	4.32	3.00	3.42
CaO	10.01	13.87	9.34	10.14	8.33	9.53	10.05	9.7	9.04	10.18	9.05	9.99	9.85	7.28	7.07	6.24	9.78
Na_2O	2.04	2.9	2.60	3.04	1.96	2.94	1.85	2.21	2.79	2.21	2.13	2.77	2.86	3.42	2.73	3.00	3.61
K_2O	1.03	0.9	1.04	1.86	1.37	1.32	2.69	0.91	0.99	1.15	1.15	0.76	1.76	1.36	1.41	1.65	1.03
P_2O_5	0.29	0.22	0.27	0.34	0.41	0.29	0.31	0.4	0.41	0.4	0.32	0.25	0.9	0.72	0.55	0.63	0.52
H_2O^+	1.46	1.50	2.47	1.19	1.8	0.76	0.79	0.78	2.61	1.29	1.66	0.30	2.1	2.32	2.08	1.45	0.97
Total	100.5	99.43	100.0	99.95	99.58	99.96	99.56	100.5	99.21	99.37	100.3	99.39	100.2	100.2	99.75	99.39	100.1
Ni	86	94.3	71	52.8	52	112	104	57	72	74	69	46	47	18	10	19	19
Co	38	40.2	36	39	44	44	40	52	39	48	48	42	29	34	33	30.6	33
Cr	108	98	83	97	89	88	100	76	44	74	54	43	96	8.8	12	9.2	17
V	253	242	264	307	375	321	375	434	234	306	346	379	204	324	285	256	205
Sc	26	26	26	26	34	31	34	37	28	35	37	60	23	26	28	23	24
Cs	0.41	0.35	0.46	0.28	–	–	–	0.39	–	0.38	0.58	1.08	1.42	0.42	0.71	0.66	0.14
Rb	19.3	19.8	18.3	26.5	36.1	23.5	29.4	19.5	25.4	23.2	28.8	18	34.3	25.3	21.2	38.6	13.6
Ba	248	243	253	312	307	259	272	202	295	305	304	221	223	434	212	461	186
Th	2.43	2.22	2.63	3.75	3.71	2.31	2.44	3.56	3.17	3.58	3.13	2.1	6.21	3.24	4.36	5.02	3.24
U	0.56	0.58	0.52	1.11	1.06	0.67	0.73	0.77	0.91	0.9	0.75	0.84	1.64	0.7	1.04	1.34	0.83
Nb	20.85	26.7	15	38.7	38.7	24.9	26.7	33.2	37.0	19.0	19.6	19.4	51.6	21.6	23.3	26.7	43.2
Ta	1.65	1.77	1.52	2.87	2.48	1.64	1.77	2.85	2.41	1.47	1.65	1.21	4.14	2.1	2.56	2.59	2.86
Sr	351	386	316	530	401	369	864	354	503	344	394	232	596	434	420	296	569
Zr	150	148	152	268	370	200	235	313	266	282	243	187	451	241	288	340	264
Hf	3.93	3.78	4.07	6.40	7.77	4.75	5.76	7.25	6.7	6.73	5.44	5.7	11.1	6.46	7.9	8.97	6.5
Y	25	22	27	32	39	30	35	46	36	45	41	40	58	45	51	54	38
Pb	3.25	2.8	3.7	25.4	–	5.2	2.9	3.0	5.1	7.6	3.0	2.6	8.8	3.1	2.2	14.2	3.5
La	19.8	21.7	17.9	36.4	34	22.5	22.6	28.3	30.2	25.9	25.4	18	56.7	30.3	36.2	41.1	34.5
Ce	43.9	48.1	39.6	82.2	78.2	51.5	52.5	62.9	70.0	62.7	56.4	41	127.3	66.9	77.6	89.1	77.6
Pr	5.38	5.86	4.89	10.2	9.88	6.71	7.05	8.35	9.19	8.17	7.09	6.0	15.8	8.37	9.7	10.9	9.86
Nd	21.8	24.8	18.8	44.2	42.4	28.6	31.5	39.6	39.8	38.2	34.4	26.0	65.0	33.8	43.5	45.6	42.9
Sm	5.19	5.36	5.01	9.24	9.14	6.48	7.35	9.74	8.86	9.03	8.11	6.3	14.5	9.26	10.3	11.2	9.54
Eu	1.74	1.82	1.66	3.03	2.93	2.04	2.23	3.06	2.95	2.67	2.63	1.85	3.51	2.65	2.88	3.27	2.82
Cd	5.30	5.16	5.43	8.59	8.83	6.35	7.36	9.86	8.55	8.74	7.51	7.2	13.2	9.65	10.89	11.39	9
Tb	0.80	0.74	0.86	1.19	1.36	0.98	1.18	1.5	1.3	1.37	1.12	1.18	1.99	1.45	1.7	1.75	1.31
Dy	4.61	4.10	5.12	7.03	7.89	5.88	7.01	8.21	7.49	7.64	6.5	7.1	11.18	8.54	9.82	10.22	7.31
Ho	0.92	0.81	1.02	1.28	1.55	1.2	1.43	1.64	1.49	1.53	1.34	1.48	2.1	1.66	1.92	2.01	1.45
Er	2.42	2.12	2.71	3.59	4.1	3.1	3.71	4.36	3.66	4.23	3.72	3.9	5.43	4.28	5.3	5.4	3.85
Tm	0.35	0.30	0.39	0.44	0.59	0.46	0.56	0.66	0.52	0.67	0.57	0.62	0.77	0.62	0.74	0.78	0.54
Yb	2.11	1.83	2.38	2.79	3.6	2.82	3.47	3.74	3.2	4.04	3.59	3.8	4.39	3.49	4.21	4.47	3.27
Lu	0.33	0.29	0.36	0.38	0.52	0.41	0.49	0.53	0.45	0.51	0.49	0.54	0.66	0.55	0.64	0.7	0.5
ΣREE	115	123.0	106.1	210.6	205.0	139.0	148.4	182.5	187.7	175.5	158.9	125.0	322.5	181.5	215.5	237.9	204.4
$(La/Yb)_n$	6.95	8.51	5.39	9.36	6.77	5.71	4.67	5.43	6.77	4.60	5.08	3.40	9.26	6.23	6.17	6.60	7.56
Eu/Eu*	1.02	1.06	0.97	1.04	1.00	0.97	0.93	0.95	1.04	0.92	1.03	0.84	0.80	0.86	0.83	0.89	0.93
Nb/Nb*	1.09	1.39	0.79	1.20	1.25	1.25	1.30	1.20	1.37	0.72	0.79	1.14	0.99	0.79	0.67	0.67	1.48

Note. Trace elements were analyzed by ICP-MS with an Elan 6100 DRC (Institute of Mineralogy, Geochemistry and Crystal Chemistry of Rare Element, D.Z. Zhuravlev), analyses 1-3,13-17 were performed with an ELEMENT-II (Institute of the Earth's Crust, Siberian Branch, Russian Academy of Sciences, S.V. Panteleeva), analysis 12 is from Polyansky (2017). $Nb/Nb^* = 0.3618 \times Nb/\sqrt{(La \cdot Th)}$, $Eu/Eu^* = Eu_n/(Sm_n \times Gd_n)^{0.5}$; n, values normalized to chondrite composition (Sun and McDonough, 1989).

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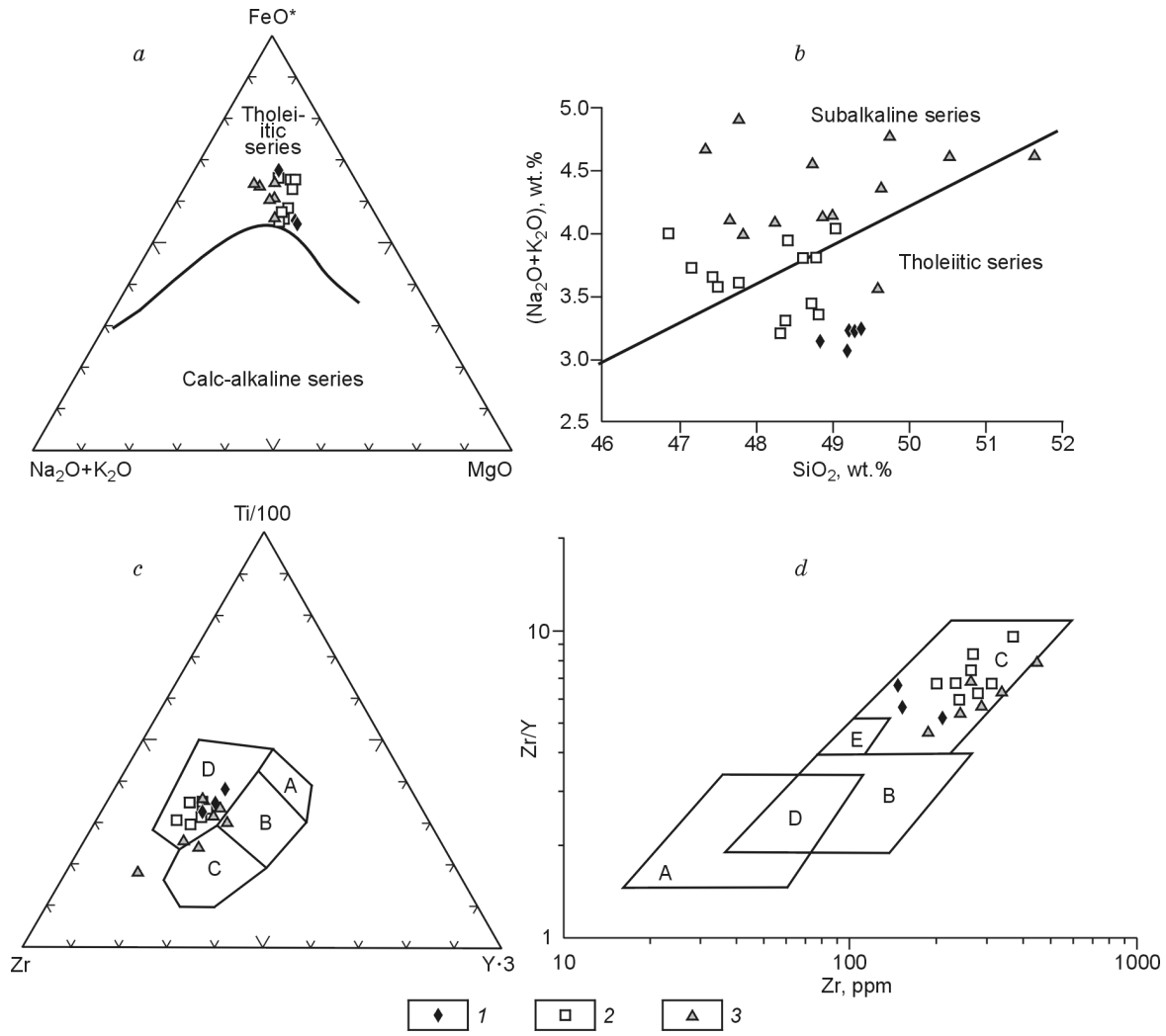


Fig. 3. Classification and discrimination diagrams for dolerites of the Vilyui paleorift. *a*, ($Na_2O + K_2O$) vs. FeO^*–MgO (Irvine and Baragar, 1971); *b*, ($Na_2O + K_2O$) vs. SiO_2 (MacDonald and Bailey, 1973); *c*, Zr–Ti–Y (Pearce and Cann, 1973) (A, island-arc tholeiites; B, MORBs; C, calc-alkaline basalts; D, within-plate basalts); *d*, Zr vs.Y–Zr (Pearce and Norry, 1979) (A, island-arc basalts; B, MORBs; C, within-plate basalts; D, island-arc basalts and MORBs; E, within-plate basalts and MORBs). *1*, dolerites of the Kontai–Dzherba zone; *2*, dolerites of the Vilyui–Markha zone; *3*, dolerites of the Chara–Sinsk zone.

we used data on dominant rock types. The basites of the Vilyui paleorift are ascribed to the tholeiitic series (Fig. 3*a*) with 46.86–51.62% SiO_2, 3.07–4.92 wt.% $Na_2O + K_2O$, and $Na_2O/K_2O > 1$. These rocks are characterized by relatively high TiO_2 (2.01–3.81%) and P_2O_5 (0.25–0.63 wt. %) content. In a SiO_2 vs. $Na_2O + K_2O$ diagram (Fig.3b), the compositions of individual intrusions within the Kontai–Dzherba zone plot within the field of quartz-normative tholeiitic basalts, based on the alkali content. The dolerites of the Vilyui–Markha dike swarm are almost equally represented by tholeiitic and subalkaline varieties. The basites of the Chara–Sinsk swarm occurs as a few dikes of common tholeiites. However, most of the intrusions are represented by high-alkaline melts, and based on their major element compositions, they fall into the field of subalkaline dolerites.

In the Zr–Ti–Y and Zr/Y–Zr discrimination diagrams (Pearce and Cann, 1973; Pearce and Norry, 1979), these rocks plot in the within-plate basalt field (Fig. 3*c*, *d*). These dolerites have a low Mg-number (34–47), indicating fractionation of a mafic melt before its ascent to the present-day chamber. In this case, the primary magmas of the Chara–Sinsk fault zone proved to be the most strongly differentiated and have the lowest Mg-number. The systematic trend toward increasing TO_2, total akalis and P_2O_5 contents is observed from the Kontai–Dzherba zone to Chara–Sinsk zone (Table 1).

Siderophile element abundances in these rocks correlate well with their Mg-numbers. The low-Mg rocks of the Chara–Sinsk dike swarm have the lowest abundances (ppm) of Ni (10), Co (29), Cr (8.8), and Sc (23). The chondrite-normalized REE patterns in these rocks (Table 2, Fig. 4*a*) show a strong REE fractionation ($(La/Yb)_n = 9.4–4.2$) and almost absent negative Eu anomalies ($Eu/Eu^* = 0.80–1.06$). They show close affinity to OIB. On spidergrams, (Fig. 4*b*), these rocks exhibit large variations in the concentrations of mobile large-ion lithophile elements Rb, Ba, Sr, which depend on a fluid phase involved. The behavior of Sr is strongly differen-

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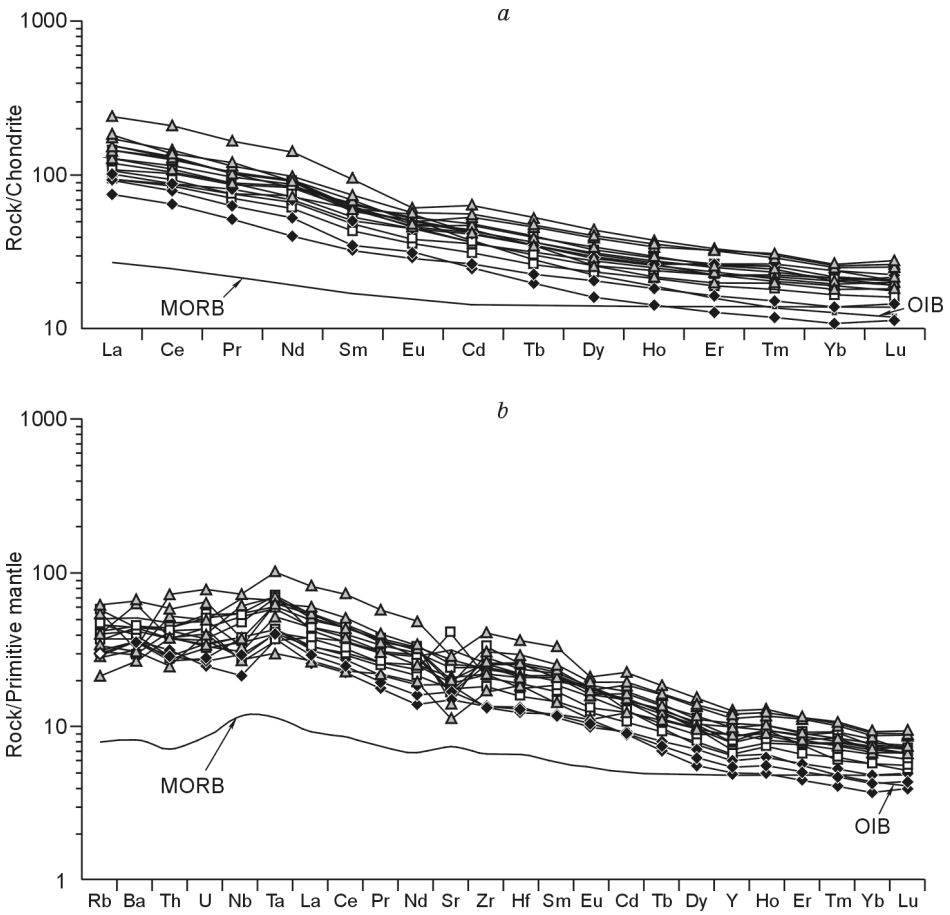


Fig. 4. Chondrite-normalized REE (*a*) and primitive mantle-normalized trace element (*b*) patterns (Sun and McDonough, 1989). Symbols are the same as in Fig. 3.

tiated, as indicated by the presence of both negative and positive Sr anomalies because of plagioclase fractionation. The highest Sr contents are characteristic of the intrusions with a high proportion of porphyritic segregations of early-formed basic plagioclase, and the lowest Sr contents are in the rocks containing more felsic varieties. The Chara–Sinsk dolerites exhibit a marked enrichment in all incompatible trace elements. The absence of negative Nb–Ta anomaly in the rocks of the Vilyui paleorift argues against significant contamination of parental melts by crustal material.

As can be seen from Tables 1 and 2 and REE and incompatible element patterns (Fig. 4*a*, *b*), the Mg and Ca contents of the parental melt systematically decrease with increasing TiO_2, P_2O_5, total alkalis, and concentrations of almost all incompatible elements from the Kontai–Dzherba to Vilyui–Markha and to Chara–Sinsk zone.

Isotope dating

On the basis of direct geological data, the dolerites of the Vilyui paleorift are estimated to be post-Silurian to Upper Permian in age. A large number of isotopic age determinations for basites have been reported by different workers in different years (Courtillot et al., 2010; Kiselev et al., 2014; Tomshin et al., 2016; Zemnukhov et al., 2005; and others). Based on the results of the analysis of all available data, we used only $^{40}Ar/^{39}Ar$ ages in this study. The previous $^{40}Ar/^{39}Ar$ dates obtained in different laboratories at different times and show very good agreement. All experiments yielded plateau ages which were weighted to ~1.5 % accuracy. The ages of dike swarms are shown in Fig. 1. The ages of dikes in the Kontai–Dzherba field are 378.0–380.7 Ma (Polyansky et al., 2017; Ricci et al., 2013). The ages of dikes in the Vilyui–Markha swarm are 368.5–376.3 Ma (Courtillot et al., 2010; Kiselev et al., 2014; Tomshin et al., 2015) (ages of 373.7–376.3 Ma predominate). The dolerites of the Chara–Sinsk swarm represent the last intrusive phase. Two dikes from the central part of the swarm yielded ages of 362.1 and 364.2 Ma (Tomshin et al., 2016). The extreme dates reported in the literature are omitted, since they require special explanation. For example, the age of 389.7 Ma (Polyansky et al., 2017) obtained from plagioclasites of the Chimidikyan swarm (north of the Vilyui–Markha swarm) can be explained by the presence of early plagioclase in the rock, defining the time of early prechamber stages of magma evolution. Younger (341.0–352.6 Ma) ages (Courtillot et al., 2010; Polyansky et al., 2017) for the Nakyn field mafic dikes (Vilyui–Markha swarm) may be due to tectonomagmatic events associated with kimberlite emplacement (Tomshin et al., 2017).

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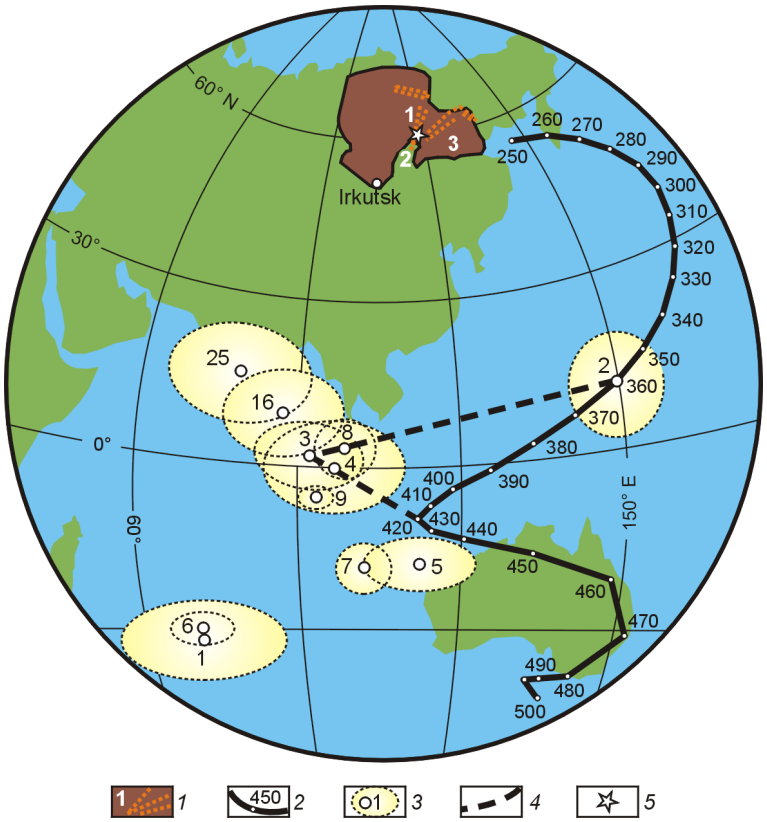


Fig. 5. Distribution of paleomagnetic poles in the zone of the dynamic influence of the Vilyui paleorift system. *1*, dike swarms (1, Vilyui–Markha; 2, Kontai–Dzherba; 3, Chara–Sinsk); *2*, apparent polar wander path of the Siberian platform, after Torsvik et al. (2012), numbers—geological age, Ma; *3*, paleomagnetic poles (the numbers are the same as in Table 3); *4*, inferred interpolation of the Devonian interval of the APWP for the Angara–Anabar block of the Siberian platform (Konstantinov et al., 2016b); *5*, Euler rotation pole (Konstantinov et al., 2011; Pavlov et al., 2008).

Therefore, based on isotope dates, we confidently assume that magmatic activity associated with the formation of the Vilyui paleorift lasted for ~18 Myr. The intrusion of basaltic melt was an uninterrupted process within the Kontai–Dzherba and Vilyui–Markha zones, whereas the intrusion of the Chara–Sinsk dikes and sills zone was interrupted by a period of rest. This period (365.5–368.8 Ma) (Ricci et al., 2013) was also marked by the emplacement of basalts of the Appainskaya and Emyaksinskaya Formations into the axial (central) part of the Vilyui paleorift.

Paleomagnetic study

Based on paleomagnetic data obtained at different sites within the Siberian Platform, Pavlov et al. (2008) and Konstantinov et al. (2011) suggested that the rift opening was associated with a relative rotation of ~20° of the Aldan block with respect to the Angara–Anabar. It was assumed that this rotation began in post-Ordovician time, and the Euler pole was located in the southwest of the paleorift: φ ~ 62° N and λ ~ 117° E (Fig. 5). At the same time, the ample information that became available in recent years, including our own age (Tomshin et al., 2016) and paleomagnetic (Konstantinov et al., 2016a,b) data on Middle Paleozoic kimberlites and mafic rocks from areas affected by the Vilyui paleorift gives more precise constraints on the beginning and extent of magmatic events under consideration.

Historically, the apparent polar wander path (Fig. 5) for the Siberian craton (Torsvik et al., 2012) was constructed mainly for units of the Angara–Anabar block. In addition, all of the paleomagnetic poles for the Middle Paleozoic (Table 3) calculated from kimberlites of the Daldyn–Alakit field, site 2 (hereinafter the site numbers are the same as in Table 3), the dolerites of the Vilyui–Markha zone, sites 2, 4, 5, basalts of the Appainskaya (D_3ap, Frasnian), site 3, and Emyaksinskaya (D_3-C_1em, Famennian–Tournaisian) Formations of the Vilyui–Markha swarm, site 2, are also attributed to this block. In connection with this, we propose to consider these sites as reference ones in geodynamic interpretation of paleomagnetic data. In contrast to them, the basites of the Chara–Sinsk and Kontai–Dzherba zones of the Aldan block have so far not been analyzed paleomagnetically.

The paleomagnetic data presented in this study were obtained during pioneering works that were carried out to the southeast of the reference sites, along the Lena River from Lensk to Sinsk, and also along its tributaries (Bol. Patom, Olekma, Sinyaya Rivers, etc.). Samples were collected in the geographic coordinate system. For sills, since they lie conformably to their host rocks, structural attitudes were used to

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Table 3. Paleomagnetic directions and poles for the zone of dynamic influence of the Vilyui paleorift

No.	Site	Coordinates of sampling locations		*N/n*	D_{av}, °	J_{av}, °	*k*, unit	α_{95}, °	Φ, °	Λ, °	*dm/dp*, °	*fm*, °	Reference
		φ, °	λ, °										
I. Angara–Anabar terrane													
Irkutsk basin													
1	Variegated silty sandstones of the Vendian Khuzhir Fm. Vhz, Irkut River	52.1	103.8	10/–	207.3	2.3	13.3	13.8	–32	71	6.9/13.8	1	Kravchinsky et al., 2001
Vilyui-Markha dike swarm													
2	Kimberlites ιPZ$_2$da, dolerites βPZ$_2$vm, basalts of the Emyaksinskaya Fm. D$_3$-C$_1$em	63.5	116.0	11/–	319.1	–49.3	31.5	8.3	11	150	8.9	31	Kravchinsky et al., 2002
3	Basalts of the Appainskaya Fm. D$_3$ap, Markha and Ygyatta Rivers	63.8	116.0	7/–	27.1	–43.8	60.3	7.8	1.9	91.7	6.1/9.7	26	Konstantinov et al., 2016b
4	Prekimberlite dolerite dikes, Mir mine βPZ$_2$vm	62.5	114.0	5/44	19.4	–44.0	53.1	10.6	–0.4	96.6	8.3/13.3	26	Konstantinov et al., 2016a
5	Prekimberlite dolerite dikes, Nyurba quarry βPZ$_2$vm	65.02	117.04	6/–	5.0	–13.4	51.8	9.4	–18.1	111.8	4.9/9.6	7	Konstantinov et al., 2016a
II. Junction zone between the Angara–Anabar and Aldan terranes													
Kontai–Dzherba dike swarm													
6	Variegated silty sandstones of the Nokhtui Fm. Vnh, Bol. Patom River	60.1	116.2	–/81	218	–16	21.5	5.2	–30	71	2.7/5.3	–8	Konstantinov, 1998
7*	High-temperature component HTC of Late Silurian rocks S$_2$lud, Nyuya River	60.6	116.3	6/56	193	16.9	11.6	20.5	–18.6	101.9	4.6	–	Powerman et al., 2013
8*	Metachronous component MTC of late Silurian rocks, Nyuya River	60.6	116.3	5/39	201.6	41.5	84.9	8.4	2.6	97.9	5.6	–	Powerman et al., 2013
9*	Metachronous component MTC of Early Silurian and Ordovician rocks, Nyuya River	60.6	116.3	8/104	204.4	37.9	27.2	2.7	–5.9	93.2	1.9/3.2	–	
10	Sill νPZ$_2$gr, Lena River, exp. 6-13	60.4	116.6	1/10	28.8 *356.0*	–51.3 *–52.4*	14.4	13.2	–	–	–	–	This study
11	Sill νPZ$_2$gr, Bol. Patom River, exp. 11-13	60.2	116.2	1/6	33.2	–55.2	10.9	21.6	–	–	–	–	
12	Sill νPZ$_2$gr, Bol. Patom River, exp. 15A-13	60.2	116.3	1/9	204.6 *17.2*	55.6 *–20.6*	19.5	12.0	–	–	–	–	
13	Sill νPZ$_2$gr, Bol. Patom River, exp. 15B-13		116.3	1/8	231.0 2.5	58.5 *–26.2*	15.2	14.7	–	–	–	–	
14	Sill νPZ$_2$gr, Bol. Patom River, exp. 16-13	60.3	116.4	1/6	12.6	–67.1	18.7	15.9	–	–	–	–	
15	Sill νPZ$_2$gr, Bol. Patom River, exp. 17-13	60.2	116.4	1/6	225.2 *32.0*	40.3 *–60.2*	16.3	17.1	–	–	–	–	
16	Sill νPZ$_2$gr, Bol. Patom River, exp. 18-13	60.1	116.5	1/6	48.8 *22.6*	–50.7 *–54.6*	26.5	13.3	–	–	–	–	
17	Average	60.2	116.3	7/51	36.4 *15.4*	–54.7 *–48.9*	52.4 *17.0*	8.4 *15.0*	10.0 –	86.6 –	8.4/11.9 –	35 –	
III. Aldan terrane													
Chara–Sinsk dike swarm (eastern flank)													
18	Dike νβPZ$_2$čs, Lena River, exp. 41-13	61.1	126.8	1/13	48.9	–54.9	70.2	5.0	–	–	–	–	
19	Silty sandstones of the Vendian Nokhtui Fm. Vnh near baked contact, Lena River, exp. 41-13	61.1	126.8	1/13	52.5	–41.9	32.5	7.4	–	–	–	–	

(*continued on next page*)

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Table 3 (*continued*)

No.	Site	Coordinates of sampling locations		*N/n*	D_{av}, °	J_{av}, °	*k*, unit	α_{95}, °	Φ, °	Λ, °	*dm/dp*, °	*fm*, °	Reference
		φ, °	λ, °										
20	Dike νβPZ$_2$čs, Lena River, exp. 42-13	61.1	126.8	1/10	46.4	–51.2	33.4	8.5	–	–	–	–	This study
21	Dike νβPZ$_2$čs, Lena River, exp. 43-13	61.1	126.8	1/7	54.1	–59.4	31.7	13.2	–	–	–	–	
22	Dike νβPZ$_2$čs, Lena River, exp. 44-13	61.2	127.8	1/7	43.2	–45.2	62.4	7.7	–	–	–	–	
23	Dike νβPZ$_2$čs, Lena River, exp. 40A-13	60.7	124.5	1/10	261.4	56.8	34.8	8.3	–	–	–	–	
24	Xenolith of granite–gneiss basement enclosed in dike, Tuolba River, exp. 40B-13	60.7	124.5	1/8	268.8	51.2	26.9	10.9	–	–	–	–	
25	Mean	61.0	126.0	7/58	58.5	–52.7	40.8	9.6	15.6	78.3	9.1/13.2	33	

Note. *N/n*, Number of sites/samples. Parameters for grouping the directions of characteristic components of NRM: D_{av}, declination; J_{av}, inclination; *k*, clustering; α_{95}, confidence oval; normal font/italics, stratigraphic/*geographical* system of coordinates. Paleomagnetic pole: Φ, latitude; Λ, longitude; *dm/dp*, confidence intervals; *fm*, paleolatitude.

* Pole coordinates not corrected for rotation of the Aldan block of the Siberian platform are accepted.

recalculate the present-day directions of the characteristic natural remanent magnetization (ChRM) vectors were recalculated into the stratigraphic coordinate system. For the dike complex, no tectonic correction was applied to the ChRM vectors.

The examples of laboratory stepwise thermal and alternating field demagnetization are shown in Fig. 6. The paleomagnetic data were processed using modern statistical graphical and analytical methods (both domestic and international). Historical approaches were considered in detail by Konstantinov (2005). The ChRM vectors were determined using principal component analysis for almost all sites (Table 3).

A positive "fold" test provides a strong evidence that the ChRM for the dolerite sills is a primary NRM since the clustering is better in the stratigraphic than in geographical system of coordinates: $k_s = 52.4 > k_g = 17.0$ (Table 3). The reversal test is also positive: the angle of the antipodal clusters of the ChRM directions is lower than the critical value: $\gamma = 6.2 < \gamma_g = 18.3$. The paleomagnetic pole for site 16 calculated from sills is in good agreement with the poles for sites 3, 4, 8 and 9 within confidence limits (Fig. 5). It should be noted that the pole of the Vendian Nokhtui Formation (site 6) of the Togus-Dabaan trough coincides with the Vendian pole (site 1) of the Khuzhir Formation of the Irkutsk basin, and the Late Silurian (Ludlow, S_2lud) pole at the Nyuya River (site 7) coincides with the poles of prekimberlite dolerite dikes (site 5) of the Nakyn field. These results suggest that there has been no net rotation of the Vilyui paleorift with respect to the Aldan block before the diking event, which is not consistent with the previous conclusions of Powerman et al. (2013).

The different situation is observed with the paleomagnetic poles measured at sites along the eastern flank of the rift. The sampled dikes (sites 18, 20–22) have ChRMs of reversed polarity, the only exception being the Tuolba dike (site 23) that recorded the normal polarity (Table 3). In this situation, the reversal test has no statistical significance. On the other hand, the ChRM directions measured in silty sandstones of Vendian Nokhtui Formation near the baked contact zone (site 19) are similar to the paleomagnetic directions of the dikes crosscutting them (sites 18, 20–22) but differ from those of the Vendian reference sites at the Khuzhir (sites 1) and Nokhtui (site 6) Formations. In the Tuolba dike, the CHRM directions of the dolerite (sites 23) and the enclosed granitogneiss xenoblock (site 24) are also identical. These data should be regarded as a positive baked contact test, which proves that the ChRM is primary. The pole at site 25 calculated for all of these sites is located far northwest of the reference sites 3, 4, 8 and 9 (Fig. 5). Therefore, the paleomagnetic data on the Chara–Sinsk dike swarm clearly indicate a relative rotation of about 22° of the Anabar–Angara block with respect to the Aldan block at the end of the Late Devonian.

Discussion and conclusions

The causes and conditions of mantle magmatism associated with the formation of the Vilyui paleorift were controlled by the plume-lithosphere interaction. The source for plume-induced magmatism was melting of a relatively homogeneous mantle with the isotopic and geochemical signatures close to those of PREMA (Kiselev et al., 2005), and OIB. It was assumed that a high-temperature mantle plume, with a diameter of about 100 km, rose beneath the craton lithosphere (Polyansky et al., 2017). The partially molten material of the mantle lithosphere (Grachev, 1977) ascended in the form of a diapir. Underplating of magmas may have been promoted by the Moho boundary acting as a high-density barrier for rising melts.

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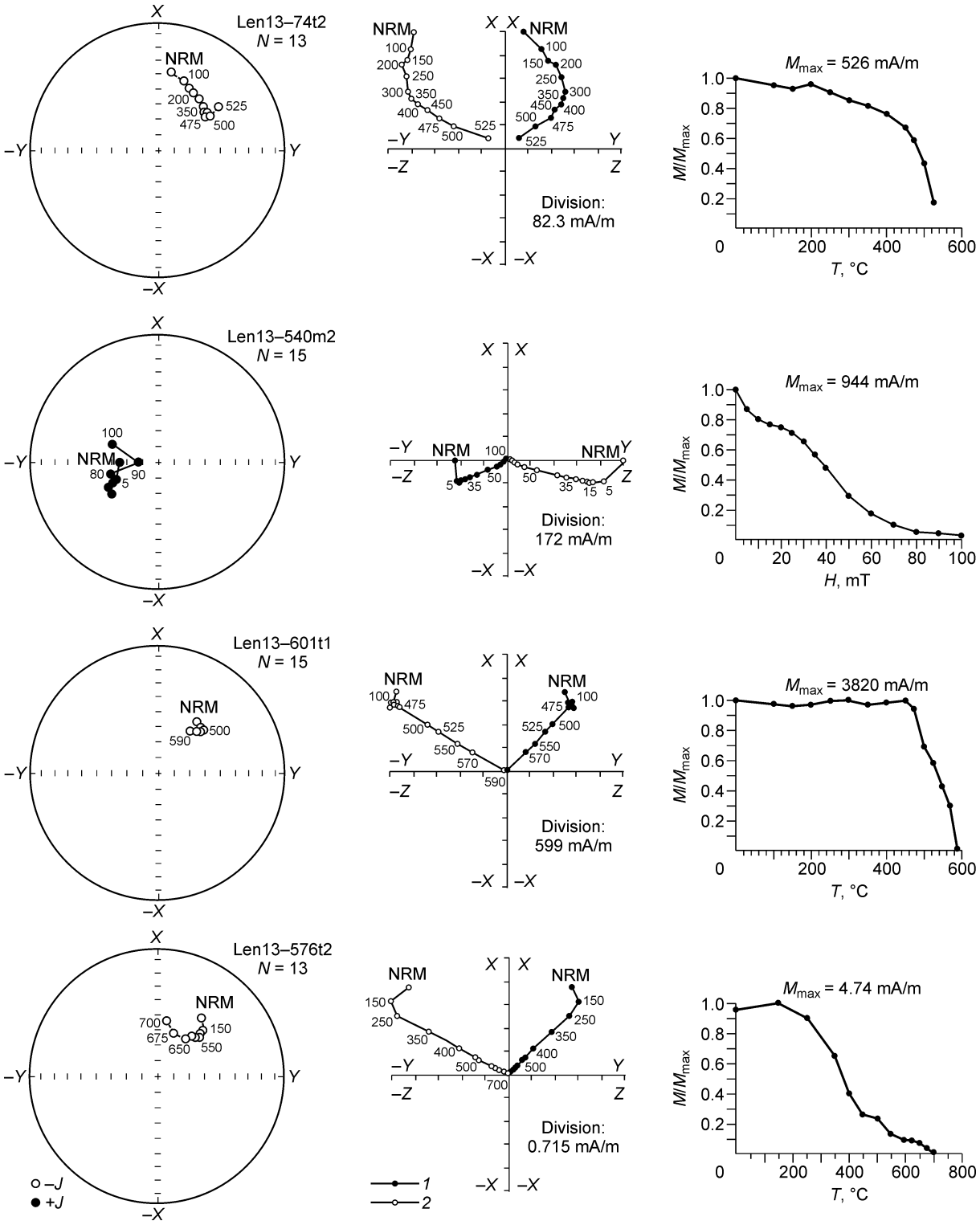


Fig. 6. Results of thermal and AF demagnetization experiments. Len13-74t2, dolerite, exposure 6, Dzherba sill; Len13-540m2, dolerite, exposure 40, Tuolba dike; Len13-601t1, dolerite, exposure 42, Ust'-Sinsk dike; Len13-576t2, silty sandstone, Nokhtui Formation near the baked contact zone, exposure 41. The numbers indicate thermal and alternating field values. Projections of NRM onto the horizontal (*1*) and vertical (*2*) planes.

The formation and accumulation of large volumes of melt at the crust–mantle boundary and subsequent spreading of magma beneath the base of the crust promoted crustal extention (rifting). Brittle failure of the crust gave rise to the zones of weakness, and the development of magma feeding conduits that channeled magmatic melts to the upper structural levels. A series of thick, multilayered sills, or more rarely, dikes were emplaced in the Kontai–Dzherba zone. The accumulation of large volumes of mafic magma at the crust–mantle boundary marks the earliest stages of crystal-

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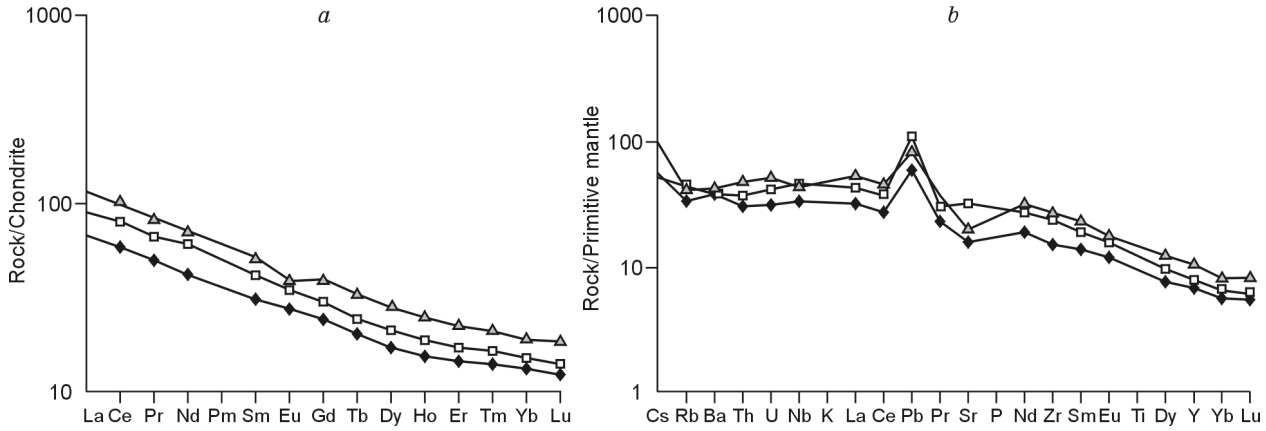


Fig. 7. Average chondrite-normalized REE (*a*) and primitive mantle-normalized trace element (*b*) patterns (Sun and McDonough, 1989). Symbols are the same as in Fig. 3.

lization of this dike swarm, particularly, the crystallization of early plagioclase, which is present in almost all varieties of gabbro-dolerites of the Vilyui paleorift. Abyssal crystallization of the mafic melt in the platform setting was discussed in detail by Oleinikov and Tomshin (1976) and Oleinikov et al. (1980). The results of the homogenization experiments on plagioclase-hosted melt inclusions showed that crystallization of the early plagioclase occurred at temperatures of 1450–1470 °C (Tomshin and Pankov, 1979) under reducing conditions at a pressure close to 12 kbar at depths of 35–40 km from the paleosurface. Under these conditions, pyrope–almandine garnets, chromian picotite, kyanite, corundum, moissanite, and a large group of native metals and intermetallic compounds crystallize simultaneously with the early anorthite–bitownite and chrysolite–forsterite (Oleinikov et al., 1980). The outpouring of large volumes of mafic magma eventually leads to collapses of magma conduits. Further melting of new batches of material, including restites, occurs at depths of magma generation corresponding to the lithospheric mantle. The melt becomes enriched in TiO_2, P_2O_5, alkalis and a set of incompatible trace elements (Tables 1 and 2, Fig. 6). The new lens of the melt rising at the Moho boundary spreads and thins westward. Simultaneously, the slip of the western flank of the Vilyui structure occurs in the same direction. This fact can reasonably explain a greater degree of crustal thinning in the eastern part of the rift with respect to its western part (Pavlov et al., 2008). The once existing local area of magma conduits begins to expand to the north of the Kontai–Dzherba structure. This results in the inception of a narrow zone of deep faults, which acted as conduits for the rising mafic magma in the form of gabbro-dolerite dikes and sills within the 373.1–376.3 Ma Vilyui–Markha swarm. The northern flank of the Vilyui–Markha zone appears to have been rotated, with a presumed time lag, eastwards to form the Chimidikyan swarm of en-echelon dikes.

The third phase of magmatic events associated with the Vilyui rift formation begins with a significant time lag (~8 Myr). The ascent and underplating of the next batch of magma beneath the crust leads to the intensive phase of crustal extension (rifting). This time (early Famennian) was the onset of the formation of the Vilyui paleorift valley. Paleomagnetic data indicate that this event was accompanied by a relative rotation of the Anabar–Angara block with respect to the Aldan block. This period was also marked by lithospheric extension and thinning, and the amount of extension of the mantle is greater than that of the crust (Polyansky et al., 2013). In the southern part of the rift, directly above the plume, a new extensive (290 km across) zone of fractures was formed, which subsequently channeled the mafic magma during emplacement of dikes of the Chara–Sinsk swarm (362.1–364.2 Ma). In these dikes, the melts have evolved toward more Ti-, P-, and alkali-rich compositions with higher REE, HFSE, and LILE contents as a result of partial melting of more refractory mantle sources.

The authors are grateful to reviewer A.E. Izokh for helpful remarks and comments.

This study was conducted as part of the research activity of the Diamond and Precious Metal Geology Institute and was supported by the Russian Foundation for Basic Research.

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